

DECEMBER 2024

Seeking Science VOL 40

SEEKING SCIENCE

MONTHLY NEWSLETTER

STEM Action Teen Institution



**The Origins of
Mathematics**
Darren Wang

**Hardware
Optimization
Through Artificial
Intelligence**

Number Theory
Kenny Wu

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The Capabilities of Biotechnology in Medicine

Richard Wang

One might wonder how humans were ever capable of developing such amazing things like CRISPR or mRNA technology? Well, these breakthroughs result from the rapidly evolving field of biotechnology. At a swift pace, it has become revolutionizing in multiple industries including medicine, agriculture, chemicals and materials, and energy. Biotechnology unlocks endless possibilities for scientists to invent many things that weren't even thought possible in the past. In the future, Biotechnology will play a big role in medicine when it comes to gene editing and regenerative medicine.

Gene editing has become one of the most revolutionary advancements in medicine due to biotechnology. Gene editing consists of CRISPR (clustered regularly interspaced short palindromic repeats) and other gene-editing tools. These technologies are capable of allowing precise and specific changes to one's DNA. Doctors have the potential to use this groundbreaking technology to cure certain genetic diseases safely. Common genetic disorders that can be treated are disorders like sickle cell anemia and cystic fibrosis. It will become



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revolutionizing to be able to treat the root causes of these genetic disorders rather than relieve the symptoms of them. However, ethical issues surround this technology due to its potential to do more harm than good. The significant leap in biotechnology has allowed for the manifestation of gene editing.

Another area of medicine, Regenerative medicine, has been unlocked by biotechnology. Regenerative medicine includes technologies such as stem cell therapies and tissue engineering. Simply put, it allows for the possible restoration or replacement of damaged tissue and organs. As a result, there is a reduced reliance on organ donors and shorter wait list times for organs. Advances in lab-grown organs could allow for life-saving treatments to take place much quicker instead of having to wait for a donor. In addition, organs damaged from other diseases such as Parkinson's or heart disease are capable of being repaired. Regenerative medicine like gene editing aims to alter the modern day methodology by treating root causes instead of the symptoms.



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Order in Chaos: Laminar Flow

Arthur Liang

Today, people are likely to be familiar with the unpredictability of fluids. Thus, they are often shocked when presented with the image of a liquid flowing so smoothly that it appears almost stationary. This is the phenomenon known as laminar flow or streamline flow, distinct from its more common, unpredictable counterpart, called turbulent flow.

Laminar flow occurs when every fluid particle follows the exact same trajectory as its preceding particle. The fluid flows in constant parallel layers with no disruptions such as eddies, swirls, or currents in the flow. This is most likely to occur when the fluid is moving relatively slowly, through a small channel, or with strong viscosity. Essentially, the viscous forces between particles must be stronger than inertial forces.

To express the ratio between viscous and inertial forces, scientists came up with a dimensionless measurement known as Reynold's number. It was proposed as the parameter in order to identify at what ratio a laminar flow will begin transitioning into a turbulent flow. Mathematically, Reynold's number is defined as $Re = \frac{\rho u d}{\mu}$, where ρ is the fluid density, u is the macroscopic velocity, d is the characteristic length, μ is the dynamic viscosity, and ν is the kinematic viscosity. At low values of Re , the flow is laminar. After a certain threshold, the flow begins transitioning towards turbulence, creating a semi-developed transition region for a specific range of Re values. Beyond this transition range, the flow becomes fully turbulent.

Laminar flow has many potential applications in academia and industry. Many laminar flows serve as the benchmark for the development of advanced simulation techniques. In industry, laminar flow exists in flows with low velocity, low density, or high viscosity, as is the case with natural convection and ventilation systems. Overall, it is obvious to see that laminar flow is a fascinating yet promising phenomenon, providing humans with a starting point to explore the unpredictable world of fluid simulations.

Science Behind Boredom

Audrey Wu

What is boredom? We have all experienced this monotonous feeling of doing nothing important. Boredom is the ongoing longing to do something without being able to. This emotion may seem pointless, yet it actually led Albert Einstein to come up with many of his genius ideas. The feeling of boredom is the negative wanting for something. There are many causes of boredom, including lack of stimulation, repetitive tasks, and perception of importance.

We have all been in those times when we're stuck in class wondering why we're learning certain things. In short, we feel bored. This boredom often stems from a lack of purpose. When you don't know why you're doing something, you feel bored. One way to combat this is by asking yourself why you're engaged in that activity.

Repetition of an action can also lead to boredom. Imagine if you really enjoy something like playing an instrument or solving math problems and practice a lot. However, after some time, you might find yourself losing passion. Repetitive practice can lead to boredom, making you lose that original spark. Therefore, even activities that were once enjoyable can become tedious through repetition.

Finally, the perception of value plays a key role in boredom. If you view an activity as unimportant, you are less likely to want to do it, and when you do, you won't be interested. To avoid boredom, remind yourself why you are doing something and consider what would happen if you didn't do it. By shifting your perspective, you can find more value in your activities and reduce feelings of boredom.

In conclusion, boredom is caused by a lack of desire to do something, repetition, and perception of importance. The feeling of boredom is the negative wanting for something. Next time you feel bored, remind yourself why you are bored and try to find a new perspective or purpose to move on.

Revolutionizing Computer Hardware Optimization Through Artificial Intelligence

Eddie Zhang

Artificial intelligence (AI) has revolutionized various industries, and its integration into computer hardware optimization has become increasingly significant. The synergy between AI and hardware design allows engineers to enhance performance, reduce costs, and innovate more efficiently. By leveraging AI techniques such as machine learning and deep learning, creating and optimizing hardware systems has entered a new era.

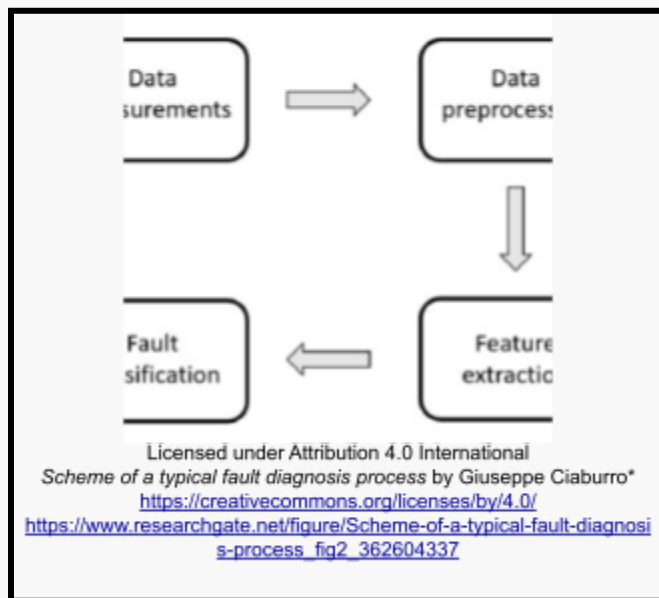
One of the primary applications of AI in hardware optimization is in chip design. Traditional microprocessors and integrated circuits design methods are time-consuming and often require extensive manual effort. AI algorithms can automate key aspects of this process, such as component placement, routing, and testing. Tools powered by AI can predict optimal configurations for complex systems, significantly reducing the design cycle while ensuring reliability and efficiency. This approach not only accelerates the development timeline but also minimizes human error.

AI also plays a crucial role in performance optimization for existing hardware. Techniques like predictive modeling and real-time monitoring allow systems to adapt dynamically to varying workloads. For instance, AI algorithms can adjust power consumption, clock speeds, and memory



allocation based on current demands, leading to improved energy efficiency and reduced operational costs. Such adaptive mechanisms are particularly beneficial in data centers, where power consumption is a critical concern.

Furthermore, AI aids in fault detection and predictive maintenance for hardware



components. By analyzing historical data and identifying patterns, AI can forecast potential hardware failures before they occur, enabling proactive repairs and reducing downtime. This capability is vital for industries that rely on uninterrupted operations, such as telecommunications, finance, and healthcare.

In addition to enhancing individual hardware components, AI is

instrumental in optimizing larger

systems like networks and distributed computing environments. Machine learning models can allocate resources more effectively across systems, ensuring maximum throughput and minimal latency. This optimization is particularly relevant for emerging technologies such as 5G and edge computing, where efficiency and speed are paramount.

The integration of AI into hardware optimization is transforming the field by enabling smarter, faster, and more sustainable solutions. As AI technologies continue to advance, their applications in hardware design and management will likely expand, paving the way for breakthroughs in computing power and efficiency. By embracing this fusion, engineers can overcome traditional limitations and shape the future of technology.

Faraday's Law of Induction

Owen Chen

The event of electromagnetic induction was dated back in 1831, observed by the English scientist Michael Faraday. In short, the discovery of the law of induction is measured as a quantity element, with the oscillation of magnetic field value having a direct proportional relationship with the electrical field within a conductor.

The visualization of the magnetic field can be seen as many induction lines spreading around the object, composing into one at both ends of the south and north poles. These lines under an area are known as magnetic flux, which are the key components to creating magnetic fields. Faraday's electromagnetic induction experiment was further improvised later on in the years by a Scottish physicist James Clerk Maxwell. Maxwell suggested the reverse also applies, where electrical fields can be used to generate magnetic fields. This concept was expanded: the fundamental relationship between magnetic field and electric fields can exist outside of conductors, in a drive electrical circuit, but in space where there's no presence of electric charges.

Maxwell constructed the expression, connecting the relationship between the change in magnetic flux with the induced electromotive (emf) based on Faraday's law of induction. This formula states the amount of emf induced in a circuit is directly related to the rate of change of the time the magnetic flux oscillates throughout the closed circuit in the unit weber. If the magnetic flux increases, then there will be more negative charged particles in the electric field, balancing out the previous electrical field. Vice versa, if the magnetic flux decreases, then there will be more positive charges in the circuit to return the previously lost electrons. However, if there's no movement in the magnetic flux, the electric field will cease to exist.

The Secret Lives of Insects: How They Communicate and Coordinate

Natalie Dai

In the unseen microcosm of the insect world, communication and coordination are key to basic survival and the elaboration of complex social systems. From the organized hives of honey bees to the swirling movements of locust swarms, insects interact with and flourish within environments in a range of ways. These too-often-overlooked methods reveal amazing elaboration of their lives.

One of the best-studied forms of insect communication is the honeybee's waggle dance. Honeybees do this peculiar dance to communicate to their fellow hive members the location of food. The orientation and duration of the dance provide detailed information about the direction and distance toward nectar-rich flowers. This method not only depicts spatial communication but also symbolizes the cooperative spirit of the hive, whereby individual effort leads to the benefit of all.

Chemical communication or pheromones also play a very important role in insect interactions. For instance, ants use chemical trails to communicate with other members of their colony regarding the location of food or potential danger. In fact, the signal is just strong that sometimes it is possible for an entire colony to behave as if it were a single entity, reacting within several minutes to changeable environmental situations. Termites also use pheromones to uphold their mound structures, informing workers when repairs are required or when living spaces should be expanded.

Other insects use auditory and visual signals for coordination as well. Crickets chirp to attract mates and advertise their fitness through the frequency and rhythm of their calls. Fireflies light up the night, communicating with bioluminescent flashes specific to their species to ensure successful mating in a competitive environment.

The Origins of Mathematics

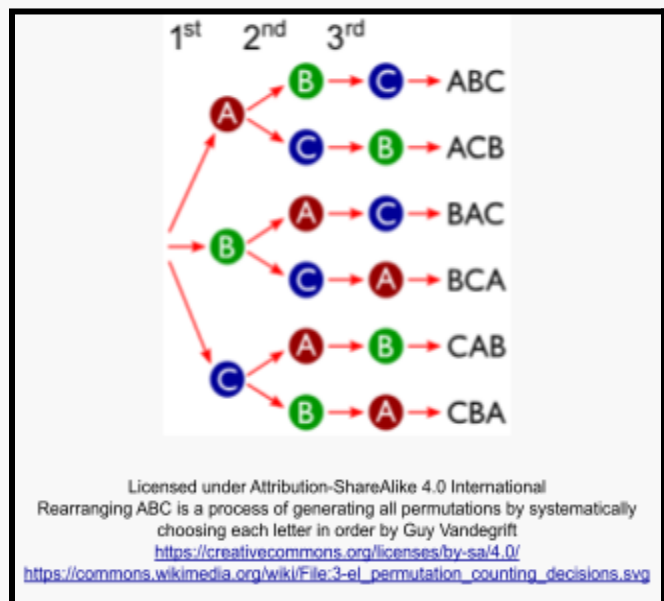
Darren Wang

Math, often called the "language of the universe," started because people needed to solve everyday problems. A long time ago, early humans would use simple marks on bones or rocks to keep track of things like how many animals they caught or how many days had passed. This was an easy way to count and remember numbers. For example, the Ishango bone, which is about 20,000 years old, shows early humans using tally marks for basic math like adding or subtracting. This shows how math began as a tool to help people with important tasks like farming, trading, hunting and sharing resources.

A long time ago, ancient civilizations like the Mesopotamians, Egyptians, and Babylonians made big improvements in math. The Babylonians came up with a number system based on 60, which is why we measure time in 60 seconds and 60 minutes and use 360 degrees for angles. The Egyptians used geometry to help them build amazing structures like the pyramids. These civilizations used math to solve real-life problems, like measuring land or predicting when the stars and planets would move. They also started writing down their math ideas, so future generations could learn from them.

However, the Greeks took math to a whole new level by making it more organized and focused on ideas rather than just solving problems. Famous thinkers like Pythagoras, Euclid, and Archimedes wanted to understand math

as something that didn't always need to be used in the real world. Euclid wrote a book



called Elements, which organized geometry into clear rules and proofs that people used for hundreds of years. Greek mathematicians also started thinking about big ideas like infinity, calculus, and patterns in numbers, showing that math could help explain the deeper truths about the universe.

After the Greek era, math grew a lot during the Islamic Golden Age. Scholars from the Islamic world saved and improved ideas from the Greeks, Indians, and Babylonians by translating their books and adding new discoveries. One important mathematician, named, Al-Khwarizmi, helped create algebra as its own field of math. At the same time, ideas from India, like the concept of zero and the decimal system, made math much easier to use for calculations. Later, these ideas spread to Europe during the Renaissance and helped scientists like Newton and Leibniz develop new fields like calculus.

As of today, math is still growing and changing, connecting ideas from the past with exciting new discoveries. It's now used for much more than

counting or measuring—it's a big part of fields like computer science, physics, and economics. Even though math started as a way to solve everyday problems, it's also became a way to explore deeper questions and understand the world better. From ancient tally marks to modern ideas like quantum mechanics, math shows how humans are always curious and creative, finding new ways to solve problems and learn about the universe.



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Number Theory

Kenny Wu

Number theory is a branch of mathematics that focuses on properties of integer and arithmetic functions. It is often referred to as “the queen of mathematics.” Attracted and fascinated mathematicians for centuries, this branch of pure mathematics captured the hearts of Ancient Greek Philosophers and modern-day scholars. Number theory revolves around understanding patterns, solving equations, unraveling the mysteries of prime numbers, and more.

To begin, number theory is a study of the structure and behavior of integers. One notable segment of the study revolves around prime numbers. A prime number is a number greater than one and has no divisors other than 1 and themselves. Prime numbers construct the fundamentals for integers with the Fundamental Theorem of Arithmetic stating that any number can be expressed as a product of primes. The randomness and endless distribution make them the talk of the town.

Another prominent number theory concept is modular arithmetic, also known as “clock arithmetic”. The structure of modular arithmetic focuses on the remainder of numbers after division by a constant modulus. Not only significant in the field of mathematics, modular arithmetic has other practical applications in cryptography, and securing digital communication. A notable example is the RSA encryption that relied on the difficulty of prime factorization of large numbers.

Furthermore, Fermat’s Last Theorem is a famous product of number theory, demonstrating the complexity and depth of the field. The theorem was proposed by Pierre de Fermat in 1637, stating that no positive integer could satisfy the equation: the sum of a to the power of n and b to the power of n equals c to the power of n for n larger than two. Despite the simplicity and straightforwardness of the equation, the theorem was left unproofed, until Andrew Wiles finally solved it in 1994.

The Process of Hearing

Denise Lee

Sound is the movement of molecules in the air that causes waves. Whenever object molecules vibrate and alter the pressure, sound comes through. The number of sound wave cycles of a particular sound is its frequency. Pitch is classified by frequency and measured in hertz (Hz). The higher the frequency, the higher the pitch, and the lower the frequency, the lower the pitch. The height of the sound wave is the amplitude. Amplitude is the loudness of the sound and its unit of measurement is decibel (dB). The brightness, darkness, heaviness, or thinness of a song is called timbre.

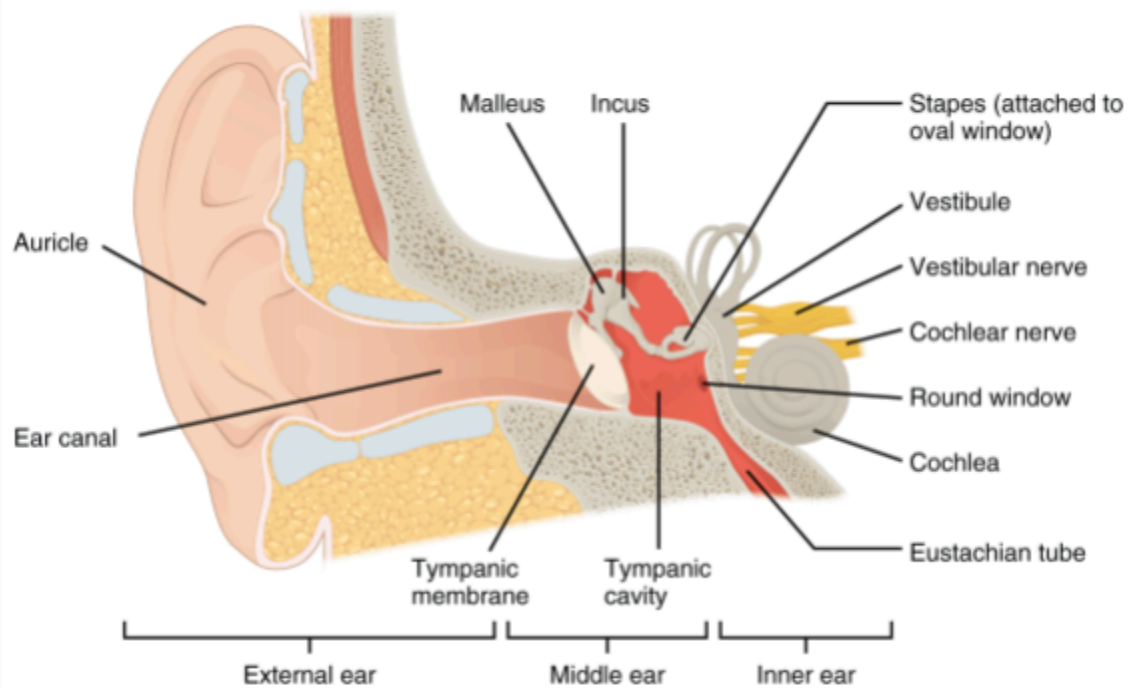
The outer ear consists of the pinna the ear canal and the ear drum. The outer part of the ear has the role of capturing sound waves and then guiding it through the ear canal where sound amplifies. The middle ear is also known as the tympanic cleft, an air-filled cavity in the ear through which sound waves travel and are transferred into the inner ear. It contains three tiny bones: The malleus, the incus, and the stapes respectively known as hammer, anvil, and stirrup. When used together they are described by the term ossicles. The ossicles join the eardrum to the oval window, which is a membrane of the inner ear. Stapes presses on the oval window, making it vibrate. The internal ear contains the cochlea and semicircular canal. The movement of the oval window causes fluids to fill the cochlea which makes the basilar membrane move.

The receptors attached to the basilar membrane are called cilia or hair cells. These hair cells move and vibrate which trigger electrical charges that are sent to the thalamus and the auditory cortex through the auditory nerve. The semicircular canal is responsible for balance. They are full of fluid and covered with calcium crystals which regulate the balance.

There are two theories that scientists came up with to explain how humans perceive pitch; place and frequency theory. Place theory states the location of hair cells

in the cochlea determines the sound frequency we hear. High-pitched sounds cause the hair cells near the base of the cochlea to vibrate, while low-pitched sounds cause hair cells near the apex to vibrate. It also states that the brain determines pitch based on which part of the cochlea is activated. Frequency theory, on the other hand, asserts that when a sound wave arrives in the ear, the whole cochlea vibrates and is translated into neural impulses. This theory believes that the frequency of the impulses defines the frequency of the sound waves. The auditory cortex is the part of the brain's temporal lobe that processes auditory information. The sound signals from the cochlea are transmitted through the auditory nerve to the thalamus and then into the auditory cortex.

When these parts of the ear work together with pitch, we can hear. Sound waves pass through our outer, middle, and inner ears, allowing us to hear different pitches and sounds.



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The Secret Lives of Insects: How They Communicate and Coordinate

Audrey Don

Insects are animals that are all around us no matter what part of earth you'd live in. They are small six-legged invertebrates with a hard outer covering and three small body parts; a head, thorax, and abdomen. These are the general features of insects, but that's as much as people would know about insects. People unless researchers would never focus their time on insect's action. For example, how the insects communicate with other insects and coordinate with other insects. Thus we have researchers referred to as Entomologists to study how insects communicate and coordinate with other insects.

Insects use a wide range of methods to share information and work together. Some rely on chemical signals called pheromones to send messages over long distances, while others use vibrations or sounds to communicate. Visual cues, such as body movements or color patterns, also play a significant role in their interactions. These various techniques allow them to coordinate complex tasks, such as building nests, finding food, or protecting their colonies.

Entomologists are scientists who specialize in studying insects, who also dedicate their efforts to unraveling these mysteries. The way insects communicate and work together is not only fascinating for others when studying but it's also critical to their survival. These methods of coordination allow them to thrive in nearly every ecosystem, from dense rainforests to bustling urban environments. Understanding how insects achieve this level of teamwork not only deepens our appreciation for these tiny creatures but also allows us to view nature's incredible efficiency. Scientists continue to uncover the secrets behind these behaviors, shedding light on a world that operates largely unseen but plays a vital role in maintaining the balance of life on Earth.

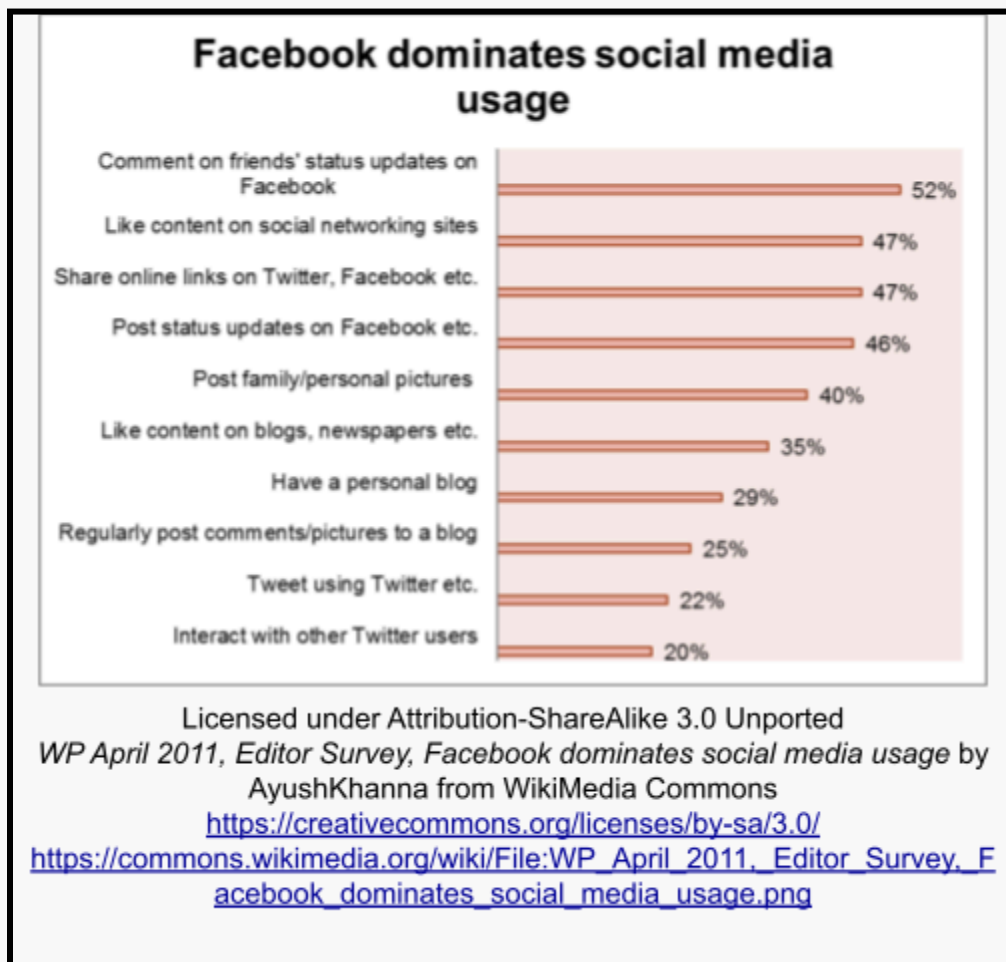
The Impact of Social Media on Human Behavior: A Psychological Perspective

Emily Ma

Social media has dramatically changed how people communicate, relate, and interact; however, it has brought considerable adjustments in human behavior, especially from a psychological perspective. One of the outstanding areas of influence concerns social comparison and self-esteem. On platforms like Instagram and TikTok, perfect renditions of other people's lives are usually displayed, encouraging comparison with often unrealistic portrayals. The phenomenon known as the "highlight reel" effect can create dissatisfaction with one's appearance, accomplishments, or lifestyle, thus exaggerating feelings of anxiety and lowering self-esteem. Some studies have found that prolonged exposure to idealized images and lifestyles increases feelings of inadequacy, especially in adolescents and young adults, and can even contribute to the development of mental health problems like depression.

Social media also breeds a unique kind of social anxiety known as the Fear of Missing Out or FOMO. The constant flow of updates can make people feel left out or excluded if they are not part of specific experiences, making FOMO a massive driver of social media addiction. It can drive people to prefer staying "in the know" online over actual world interactions and responsibilities. Research has shown that FOMO leads to anxiety and even sleep disturbances as users will ignore rest in favor of staying engaged, disconnecting themselves from their natural cycles and inner peace.

Another strong influence is the development of addictive behaviors associated with social media usage. Many platforms are designed to hook users by triggering dopamine reactions through likes, comments, and other reward mechanisms. Such behaviors like those in substance addiction, where the person seeks the dopamine "hits" delivered by notifications and interactions. Over time, this can lead to compulsive checking habits that interfere with productivity, face-to-face relationships, and mental clarity. Understanding these psychological effects is essential to enable solutions for the broader implications of social media and promote healthier behaviors when using it.



R Values and U Values

Aidan Hong

Typically, in order to maintain heat in a house, engineers will use thermodynamics in order to determine what kind of insulation should be used. While many different types of materials exist to handle this problem, certain materials are better than others. This can be determined through R values and U values

The R value is used to determine a material's ability to resist heat transfer. Typically, a high R value indicates that the material is good at resisting heat transfer, and a low R value indicates it is not very good at resisting heat transfer. When creating insulation, engineers typically aim to maximize the R value. Most of the time, engineers use more than one material to create insulation. In order to determine the total R value, you can simply add up all the R values to get the total R value.

The U value, on the other hand, determines conductivity. This means that a high U value will transfer heat very quickly, and a low U value will not transfer heat very quickly. In terms of insulation, a low u value is desirable as a high u value will transfer heat very quickly. U values and r values are related: to find the u value, you can simply take the reciprocal of the r value.

R values and u values are critical when creating insulation. Especially during summer or winter, these r values and u values are important to engineers: they tell a lot about a material's ability to resist heat, maintain heat, and more. Without these numbers, we will not be able to tell if an insulation works or not through math. These numbers save countless hours that would be otherwise spent on testing and countless dollars from unnecessary testing.

The Sardine Run

Arick Hong

Once a year, millions of sardines gather together along the coast of Africa. While the reason for this bizarre event is still unknown, the presence of so many sardines creates a feeding frenzy for many animals.

Sardines are a type of small fish. These fish usually gather as a form of self-defense, since targeting one sardine in a giant ball is very difficult. However, some animals have a trick to attack these fish. Dolphins, seals, sharks, and other animals have a strategy where they circle the group of sardines into a tight ball. The sardines cannot move out of the ball due to the presence of these predators outside of it. The predators then attack from all sides, which closes all exits for the sardines. These sardines become so desperate that they try jumping out of the water, right into the mouths of seabirds. Bryde's Whales also attack the sardines as well, gulping them with their giant mouths. The amount of sardines present creates a giant food source for these animals, which is why the run itself is so important.

Although the sardine run is important for animals, it is also important for people too. Fishermen flock to the sea to catch sardines using large nets. All of these sardines create a large food source for their people and cities. This helps their people thrive and their city flourish. While some people may think that all of this sardine consumption will affect the run itself, the amount of sardines is so high that it barely creates a dent. The run itself is actually so large it is visible from space.

Understanding Mitosis: The Key to Cellular Growth and Division

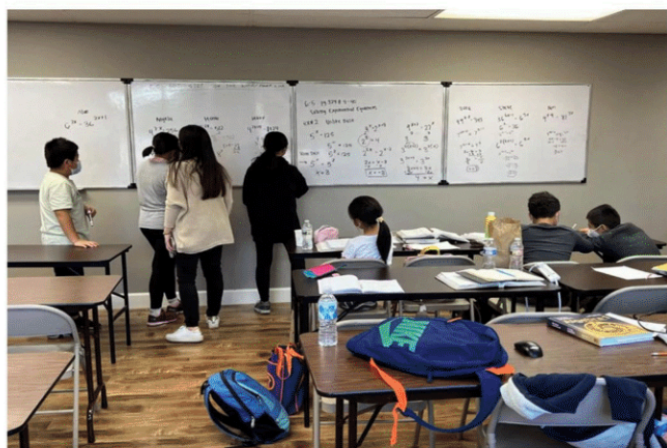
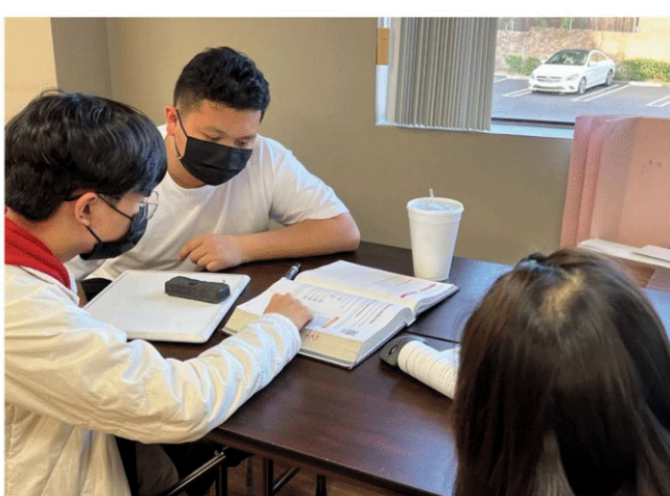
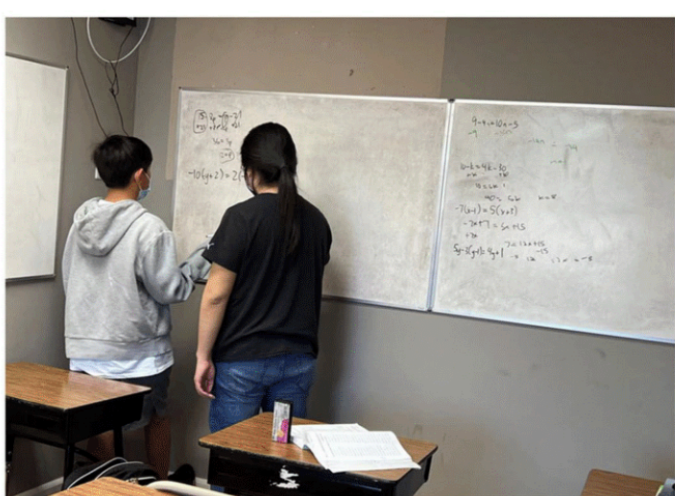
Brandon Pian

When you scrape a knee, what do you think your cells are doing to repair your skin. Your cells are actually going through a process called mitosis which allows cells to divide. Mitosis is the process by which a eukaryotic cell divides to form two genetically identical daughter cells. This division is crucial for growth, tissue repair, and asexual reproduction. Mitosis ensures that each daughter cell receives an exact copy of the parent cell's DNA.

The process of mitosis is divided into several stages: prophase, metaphase, anaphase, telophase, and cytokinesis. In prophase, the chromatin condenses into visible chromosomes, and the nuclear envelope begins to break down. The spindle fibers, which are responsible for moving the chromosomes, start to form. During metaphase, the chromosomes align along the cell's equatorial plane, known as the metaphase plate. Anaphase follows, where the sister chromatids are pulled apart toward opposite poles of the cell. By telophase, the separated chromatids reach the poles, and new nuclear membranes form around each set of chromosomes.

Finally, cytokinesis occurs, dividing the cytoplasm and completing the process. In animal cells, the membrane pinches inwards to form two separate cells, while in plant cells, a cell plate forms to create a new cell wall.

The result of mitosis is two daughter cells, each with a full set of chromosomes identical to the parent cell. Mitosis is vital for maintaining genetic stability in an organism. Any errors during mitosis, such as improper chromosome separation, can lead to diseases like cancer. Thus, the precision of mitosis is essential for healthy cell function and organismal growth.



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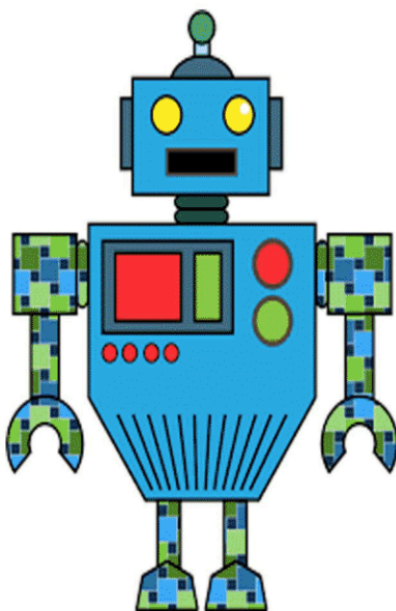
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4-12年級為AP Computer 課程準備
並可參加全國及各項國際AI競賽

SAT 英文寫作班
4-12年級, 閱讀, 文法, 寫作
** 因材施教 突破盲點 **

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2022年賽季將結束，每個小朋友都忙著完成自己的機器人組裝和編程

